



# A variant calling pipeline for rare Mendelian disease diagnosis using simulated trio-based exome sequencing data

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## Abstract

Rare Mendelian diseases affect millions of people around the world, yet many remain undiagnosed due to the complexity of their genetic basis. Trio-based exome sequencing, in which a proband (child) and both parents are sequenced together, is an effective diagnostic strategy that enables the identification of disease-causing variants through inheritance-based filtering. In this report, we implemented a bioinformatics pipeline for the analysis of simulated paired-end exome sequencing data derived from ten trios (five per student), all restricted to chromosome 20 and aligned to the GRCh38 reference genome. The pipeline included read alignment with Bowtie2, quality control with FastQC and Qualimap, variant calling with FreeBayes, and variant annotation with the Ensembl Variant Effect Predictor (VEP). Inheritance-based filtering was applied according to the mode of inheritance assigned to each trio: autosomal recessive (AR) for trios 1, 4 and 5; autosomal dominant inherited for trio 2; and autosomal dominant de novo for trio 3. Candidate disease-causing variants were identified in eight out of ten trios, including frameshift mutations in *KCNQ2* (Seizures Benign Familial Neonatal 1 / Developmental and Epileptic Encephalopathy 7), a de novo frameshift in *ASXL1* (Bohring-Opitz syndrome), a de novo frameshift in *SLC12A5* (Developmental and Epileptic Encephalopathy 34), a frameshift in *SLC2A10* (Arterial Tortuosity Syndrome), stop-gained variants in *SLC52A3* (Brown-Vialetto-Van Laere Syndrome 1) and *GSS* (Glutathione Synthetase Deficiency). Two trios did not present candidate variants and were reported as unaffected.

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## 1 Introduction

Rare diseases, defined as conditions affecting fewer than 1 in 2000 individuals, represent a major diagnostic challenge in clinical medicine. The majority of rare diseases have a genetic origin, with a significant proportion following patterns of Mendelian inheritance [1].

Next-generation sequencing (NGS), and in particular whole-exome sequencing (WES), has substantially improved rare disease diagnosis. By capturing the protein-coding regions of the genome, WES allows the simultaneous examination of thousands of genes at relatively low cost compared to whole-genome sequencing [2]. A particularly effective diagnostic strategy involves sequencing patient-parent trios, enabling the direct identification of de novo mutations and inherited variants, greatly facilitating the filtering of candidate disease-causing alleles [3].

Variant interpretation in a clinical genomics context relies on a combination of computational predictions, population frequency data, and disease-gene databases

such as OMIM and ClinVar. Variants are typically prioritized based on their predicted functional impact, rarity in the general population, and concordance with the patient's inheritance pattern and phenotype.

In this report, we present a complete diagnostic workflow applied to simulated paired-end trio-based exome sequencing data. Paired-end sequencing is a strategy in which both ends of the same DNA fragment are sequenced, producing two reads separated by a known approximate distance referred to as the insert size [4]. Ten trios derived from 1000 Genomes Project samples were analyzed (five trios per student), with disease-causing variants simulated in silico. The analysis was restricted to chromosome 20 and performed against the GRCh38 reference genome. We demonstrate the successful identification of pathogenic variants in eight out of ten trios, with diagnoses of epileptic encephalopathies, a congenital malformation syndrome, a vascular connective tissue disorder, a riboflavin transporter deficiency, and an inborn error of metabolism.

## 2 Materials and Methods

### Data

The simulated sequencing data were provided for ten trios (trios 1–5 per student), each consisting of a proband (child), father, and mother. The trios raw reads were simulated starting from real genotypes from the 1000 Genomes Project, stored in the directory `/home/BCG2026_exam/` on the leon cluster (IP: 159.149.160.7). The sequencing data were restricted to chromosome 20 and aligned against the GRCh38 reference genome. Target regions were defined by the `chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed` file. The modes of inheritance are the same for both students: autosomal recessive (AR) for trios 1, 4 and 5; autosomal dominant inherited (AD inh.; mother affected) for trio 2; and autosomal dominant de novo (AD dn) for trio 3. Each student worked on a separate cluster account. Although filenames were identical, the disease-causing variants were independently simulated and lead to different molecular diagnoses.

### Read Alignment and Quality Control

Raw reads were aligned to the GRCh38 chromosome 20 reference using Bowtie2 [5] with the default parameters for paired-end data. Read groups were assigned to each sample using the `--rg-id` and `--rg SM` flags to ensure correct sample identification during the joint variant calling. The aligned reads were then directly streamed into SAMtools to be converted into BAM format, coordinate-sorted, and indexed [6]. Quality control was performed using FastQC on all BAM files, and Qualimap [7] was run per sample using the exome target BED file to assess on-target coverage. A multi-sample QC report was generated with MultiQC [8].

### Variant Calling

Joint variant calling across all three members of each trio was performed using FreeBayes [9]. Calls were filtered using a minimum mapping quality of 20 (`-m 20`), minimum alternate allele count of 5 (`-C 5`), minimum base quality of 10 (`-Q 10`), and minimum site coverage of 10 (`--min-coverage 10`). Output VCF files were compressed with bgzip and indexed with bcftools.

## 3 Results

### Alignment and Quality Control

All thirty samples (three per trio, ten trios across the two students) achieved excellent alignment metrics, with overall alignment approaching 100.00% as reported by Bowtie2. The mean coverage, assessed by Qualimap, ranged approximately from 38x to 42x across all samples, well above the standard clinical threshold of 20x. No samples were excluded based on their quality. FastQC and Qualimap reports were aggregated in a single MultiQC report.

### Inheritance-Based Filtering

The resulting variants were filtered using bcftools according to the mode of inheritance of each trio. For AR trios (1, 4, 5), variants were retained where the child was homozygous for the alternate allele (`GT[0]="AA"`) and both parents were heterozygous (`GT[1]="RA"`, `GT[2]="RA"`). For AD inherited trio 2 (mother affected), the filter required the child and mother to be heterozygous and the father to be homozygous for the reference allele. For AD de novo trio 3, only variants where the child was heterozygous and both parents were homozygous for the reference were retained. All variants were additionally filtered for `QUAL > 20` and restricted to the exome target regions using the `-R` flag.

### Variant Annotation and Interpretation

Filtered variants were annotated using the Ensembl Variant Effect Predictor (VEP) [10] with the GRCh38 assembly, selecting one consequence per variant per MANE Select transcript (`--mane --pick-allele`). Population allele frequencies were retrieved from gnomAD exomes and 1000 Genomes. Final candidate variants were retained using `filter-vep` with the parameters: HIGH or MODERATE impact and `MAX_AF < 0.0001` or not reported. Candidate genes were cross-referenced against the provided list of disorders simulated in the dataset and against OMIM and ClinVar databases. Disease annotation was additionally verified through the VEP web interface using RefSeq transcripts.

### Coverage Visualization

Per-sample coverage tracks in BEDGraph format were generated using `bedtools genomecov` [11] with a maximum depth of 100x and uploaded to the UCSC Genome Browser for visualization of candidate variant regions.

### Reproducibility

All scripts and intermediate files (BAM, VCF, annotated VCF, QC reports, coverage tracks) are stored on the leon cluster at:

`/home/BCG2026_Matassa_F/exam/`

`/home/BCG2026_Salmin_L/exam-project/`

The complete analysis script used by both students is reported in the Data Availability section at the end of the report.

MultiQC summary plots are reported in Figures 1, 2 and 3.

| Sample Name   | % GC | Ins. size | $\geq 30X$ | Median cov | Mean cov | % Aligned | % Dups |
|---------------|------|-----------|------------|------------|----------|-----------|--------|
| trio_1_child  | 52%  | 299       | 95.0%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_1_father | 52%  | 299       | 94.8%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_1_mother | 52%  | 299       | 95.2%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_2_child  | 52%  | 300       | 94.5%      | 40.0X      | 39.7X    | 100.0%    | 2.9%   |
| trio_2_father | 52%  | 299       | 95.0%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_2_mother | 52%  | 299       | 95.2%      | 40.0X      | 40.1X    | 100.0%    | 2.9%   |
| trio_3_child  | 52%  | 300       | 94.6%      | 40.0X      | 39.9X    | 100.0%    | 2.9%   |
| trio_3_father | 52%  | 299       | 95.2%      | 40.0X      | 40.1X    | 100.0%    | 2.9%   |
| trio_3_mother | 52%  | 299       | 94.9%      | 40.0X      | 39.9X    | 100.0%    | 2.8%   |
| trio_4_child  | 52%  | 299       | 95.0%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_4_father | 52%  | 299       | 94.9%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_4_mother | 52%  | 299       | 95.0%      | 40.0X      | 40.0X    | 100.0%    | 2.9%   |
| trio_5_child  | 52%  | 299       | 95.2%      | 40.0X      | 40.1X    | 100.0%    | 2.9%   |
| trio_5_father | 52%  | 299       | 95.2%      | 40.0X      | 40.1X    | 100.0%    | 2.9%   |
| trio_5_mother | 52%  | 300       | 95.6%      | 40.0X      | 40.2X    | 100.0%    | 2.9%   |

Figure 1: MultiQC general statistics summary across Salmin samples, showing GC content, sequence length, mean and median coverage, % aligned reads and % duplicate reads. Metrics are consistent with Matassa samples.

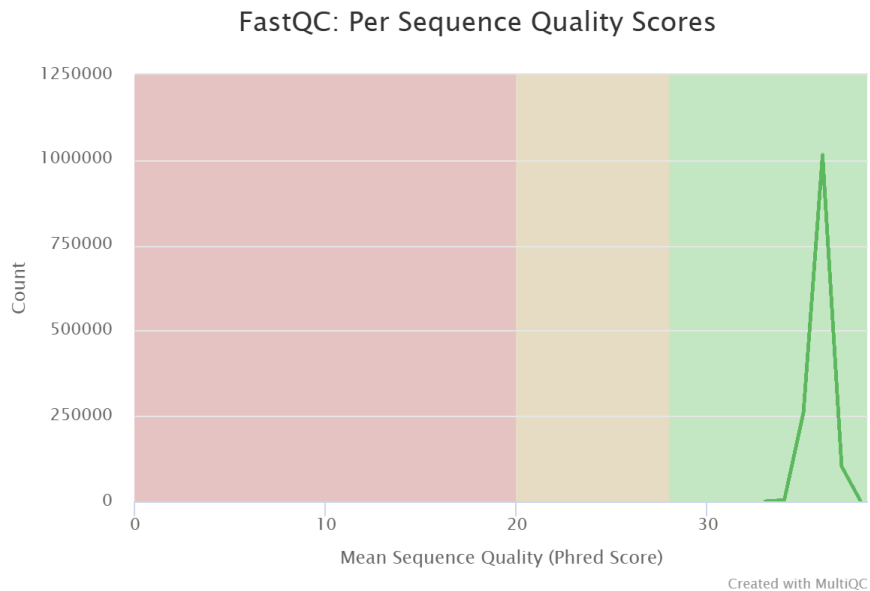


Figure 2: FastQC per-sequence quality scores aggregated across Salmin samples. Mean Phred scores are well above 30 for the vast majority of reads, indicating a high-quality sequencing.

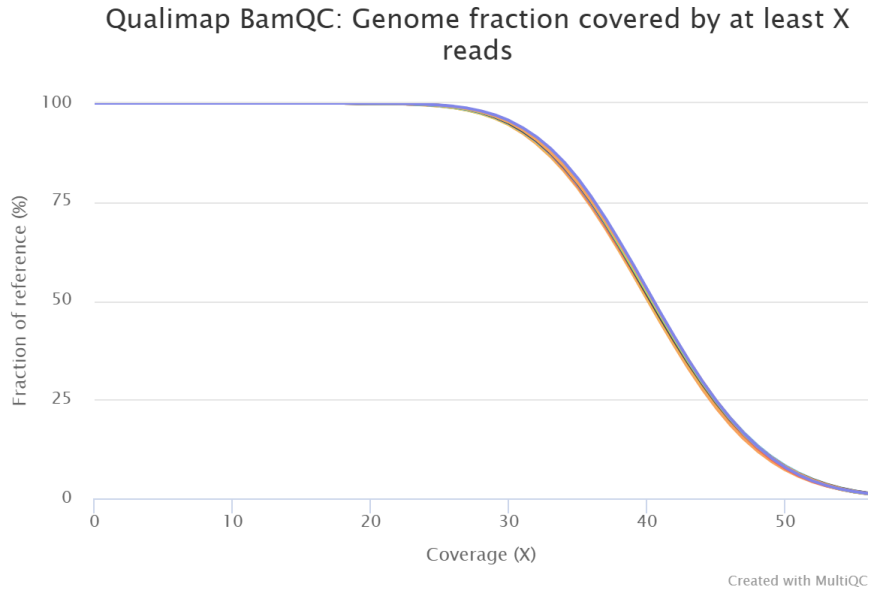


Figure 3: Qualimap genome fraction covered at increasing depth thresholds. A substantial fraction of the target regions is covered at  $\geq 30x$  in all samples between students.

## Variant Calling and Filtering

FreeBayes identified approximately 9,000–9,100 raw variant sites per trio on chromosome 20. After inheritance-based genotype filtering and quality filtering ( $QUAL > 20$ ), the number of candidate variants was substantially reduced (approximately 100–260 variants per trio depending on the inheritance mode). After VEP annotation and frequency/impact filtering (HIGH or MODERATE impact,  $MAX\_AF < 0.0001$  or not reported), one to two high-confidence candidate variants per trio were identified in affected trios; two trios (one per student) instead presented no candidate variants after passing the final filter.

## Candidate Variant Identification and Diagnosis

Candidate variants and final diagnoses for all ten trios are summarized in Table 1. In all affected trios, the predicted causative variants were either frameshift insertions/deletions or stop-gained substitutions, all with high impact according to VEP. All identified disease genes are present in the provided list of simulated disorders and are consistent with the assigned mode of inheritance. Two trios (Matassa trio 1 and Salmin trio 1) presented no high-impact rare variants according to their AR pattern and were therefore reported as unaffected, in line with the expectation that approximately 20% of cases are simulated without disease-causing variants.

Table 1: Summary of diagnostic findings for all ten trios across Matassa and Salmin.

| Student / Trio  | Mode    | Gene           | Variant type          | Genotype (C/F/M) | Diagnosis                                     |
|-----------------|---------|----------------|-----------------------|------------------|-----------------------------------------------|
| Matassa, Trio 1 | AR      | –              | –                     | –                | Unaffected                                    |
| Matassa, Trio 2 | AD inh. | <i>KCNQ2</i>   | Frameshift insertion  | 0/1, 0/0, 0/1    | Seizures Benign Familial Neonatal 1 / DEE 7   |
| Matassa, Trio 3 | AD dn   | <i>ASXL1</i>   | Frameshift insertion  | 0/1, 0/0, 0/0    | Bohring-Opitz Syndrome                        |
| Matassa, Trio 4 | AR      | <i>SLC2A10</i> | Frameshift deletion   | 1/1, 0/1, 0/1    | Arterial Tortuosity Syndrome                  |
| Matassa, Trio 5 | AR      | <i>GSS</i>     | Stop gained (p.Q308*) | 1/1, 0/1, 0/1    | Glutathione Synthetase Deficiency             |
| Salmin, Trio 1  | AR      | –              | –                     | –                | Unaffected                                    |
| Salmin, Trio 2  | AD inh. | <i>KCNQ2</i>   | Frameshift deletion   | 0/1, 0/0, 0/1    | Seizures Benign Familial Neonatal 1 / DEE 7   |
| Salmin, Trio 3  | AD dn   | <i>SLC12A5</i> | Frameshift deletion   | 0/1, 0/0, 0/0    | Developmental and Epileptic Encephalopathy 34 |
| Salmin, Trio 4  | AR      | <i>SLC52A3</i> | Stop gained (p.E249*) | 1/1, 0/1, 0/1    | Brown-Vialetto-Van Laere Syndrome 1           |
| Salmin, Trio 5  | AR      | <i>SLC52A3</i> | Stop gained (p.E249*) | 1/1, 0/1, 0/1    | Brown-Vialetto-Van Laere Syndrome 1           |

## Coverage Visualization

Coverage tracks for all samples were generated in BED-Graph format and uploaded to the UCSC Genome Browser (GRCh38/hg38). As an example, Figure 4 shows coverage tracks for Matassa trio 5 at the *GSS* locus (chr20:34,932,046), where the child displays homozygous coverage of the alternate allele and both parents display heterozygous coverage, consistent with the AR inheritance pattern.

## 4 Discussion

We successfully implemented a complete diagnostic genomics pipeline for trio-based exome sequencing data, identifying pathogenic variants in eight out of ten trios. The pipeline correctly identified HIGH impact variants — frameshifts and stop-gained substitutions — as the primary disease-causing events. All identified disease genes were consistent with the assigned mode of inheritance and matched entries in the simulated disorder list, OMIM [12] and ClinVar [13].

The two *KCNQ2* frameshift variants (OMIM #121200, OMIM #613720) identified in trio 2 of both students represent well-characterized causes of neonatal epilepsy, with dominant inheritance and high penetrance. The identification of the causative variant in the affected mother in both cases confirms the AD inherited pattern. The de novo frameshifts in *ASXL1* (OMIM #605039, Matassa trio 3) and *SLC12A5* (OMIM #616645, Salmin trio 3) are consistent with the known mutational mechanisms of Bohring-Opitz syndrome and DEE 34, respectively. Bohring-Opitz syndrome is predominantly caused by de novo *ASXL1* mutations [14], whereas *SLC12A5*-related encephalopathy is often described with biallelic variants [15]. The AR cases — *SLC2A10* (OMIM #208050, Matassa trio 4), *GSS* (OMIM #266130, Matassa trio 5), and *SLC52A3* (OMIM #211530, Salmin trio 4 and trio 5) — all follow classic AR inheritance, with the child homozygous and both parents heterozygous carriers. Notably, this same *SLC52A3* variant was identified in both AR trios.

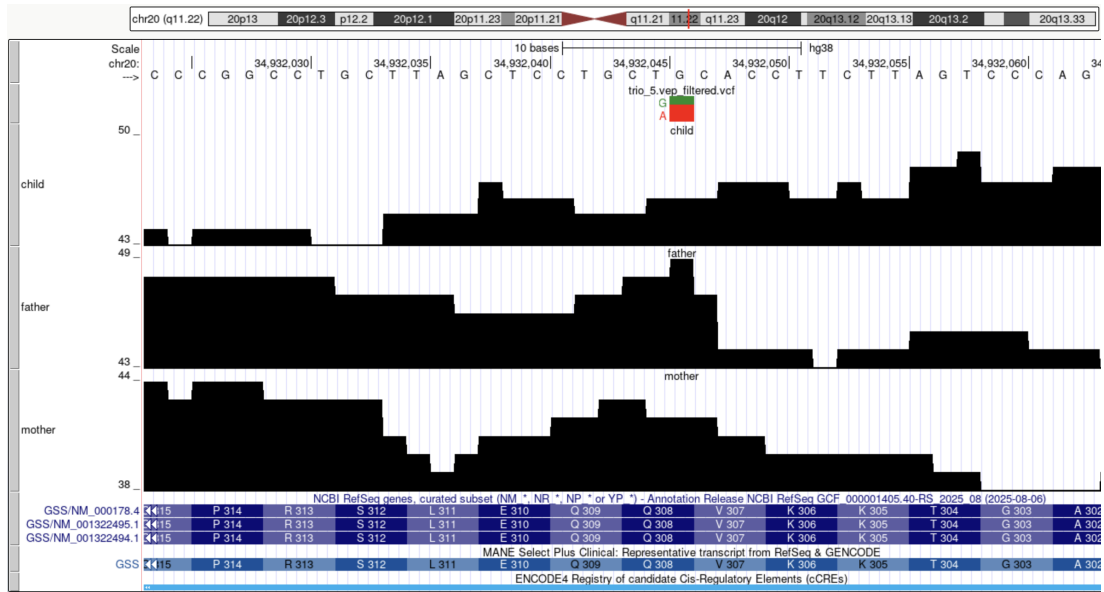
Both students reported unaffected probands in their respective trio 1, consistent with the expected ~20% of cases simulated without disease-causing variants; the pipeline successfully achieved a diagnostic yield of 80%, demonstrating its effectiveness for trio-based rare disease analysis on a controlled dataset.

## 5 References

### References

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Figure 4: UCSC Genome Browser screenshot showing coverage tracks for Matassa trio 5 at the *GSS* locus (chr20:34,932,046). The child (top) shows homozygous alternate genotype at the variant position with consistent coverage across the locus ( $\sim 43\text{--}50\times$ ).



## Data Availability

The complete pipeline used by both students is reported below. All intermediate files (BAMs, VCFs, annotated VCFs, QC reports and coverage tracks) are stored in the corresponding home directories on the leon cluster.

```
#!/bin/bash

for trio in trio_1 trio_2 trio_3 trio_4 trio_5; do
  cd $trio
  echo "=== Processing $trio ==="

  # --- Read alignment ---
  bowtie2 -1 HG00406.targets_R1.fq.gz -2 HG00406.targets_R2.fq.gz \
    -x /home/BCG2026_exam/chr20 \
    --rg-id "child" --rg "SM:child" \
    | samtools view -Sb | samtools sort -o ${trio}_child.bam
  bowtie2 -1 HG00407.targets_R1.fq.gz -2 HG00407.targets_R2.fq.gz \
    -x /home/BCG2026_exam/chr20 \
    --rg-id "father" --rg "SM:father" \
    | samtools view -Sb | samtools sort -o ${trio}_father.bam
  bowtie2 -1 HG00408.targets_R1.fq.gz -2 HG00408.targets_R2.fq.gz \
    -x /home/BCG2026_exam/chr20 \
    --rg-id "mother" --rg "SM:mother" \
    | samtools view -Sb | samtools sort -o ${trio}_mother.bam

  # --- BAM indexing ---
  samtools index ${trio}_child.bam
  samtools index ${trio}_father.bam
  samtools index ${trio}_mother.bam

  # --- Quality control ---
  fastqc *.bam
  qualimap bamqc -bam ${trio}_child.bam \
    --feature-file ../chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed \
    -outdir ${trio}_child
  qualimap bamqc -bam ${trio}_father.bam \
    --feature-file ../chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed \
    -outdir ${trio}_father
  qualimap bamqc -bam ${trio}_mother.bam \
    --feature-file ../chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed \
    -outdir ${trio}_mother

  # --- Variant calling ---
  freebayes -f /home/BCG2026_exam/chr20.fa \
    -m 20 -C 5 -Q 10 --min-coverage 10 \
    ${trio}_child.bam ${trio}_father.bam ${trio}_mother.bam \
    > ${trio}.vcf
  bgzip ${trio}.vcf
  bcftools index ${trio}.vcf.gz
done
```

```

# --- Inheritance-based filtering ---
if [ $trio == "trio_1" ] || [ $trio == "trio_4" ] || [ $trio == "trio_5" ]; then
    FILTER='GT[0]="AA" && GT[1]="RA" && GT[2]="RA'
elif [ $trio == "trio_2" ]; then
    FILTER='GT[0]="RA" && GT[1]="RR" && GT[2]="RA'
elif [ $trio == "trio_3" ]; then
    FILTER='GT[0]="RA" && GT[1]="RR" && GT[2]="RR'
fi
bcftools view -R ../chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed ${trio}.vcf.gz \
| bcftools view -S ../samples.txt | bcftools view -i "$FILTER" \
| bcftools filter -i 'QUAL>20' -Ov -o ${trio}.cand.vcf

# --- Annotation with VEP ---
vep -i ${trio}.cand.vcf -o ${trio}.vep_annotated.vcf \
--vcf --cache --offline --assembly GRCh38 \
--dir_cache /data/vep_cache --use_given_ref --mane \
--pick_allele --af --af_1kg --af_gnomade --max_af \
--sift b --polyphen b --no_fasta

# --- Final filtering ---
filter_vep -i ${trio}.vep_annotated.vcf \
-o ${trio}.vep_filtered.vcf \
--filter "(IMPACT is HIGH or IMPACT is MODERATE) and (not MAX_AF or MAX_AF < 0.0001)"

# --- Coverage tracks ---
bedtools genomecov -ibam ${trio}_child.bam -bg -trackline \
-trackopts 'name="child"' -max 100 > ${trio}_childCov.bg
bedtools genomecov -ibam ${trio}_father.bam -bg -trackline \
-trackopts 'name="father"' -max 100 > ${trio}_fatherCov.bg
bedtools genomecov -ibam ${trio}_mother.bam -bg -trackline \
-trackopts 'name="mother"' -max 100 > ${trio}_motherCov.bg

cd ..
done

# --- Aggregate QC report ---
multiqc trio_*/

```